

EEE3095/6S Class Test

October 2024

Duration: 75 minutes

Total: 50 marks

Empl ID: _____

Instructions:

- This is a closed-book test.
- There are **two** Sections (A and B) to this test.
- You must use a pen to complete the test – pencil will only be marked for drawings. Values indicated on drawings must be written in pen.
- Please show all your working clearly – your method and approach matter.
- Keep all your answers within the allocated blocks; anything outside of these blocks will not be marked.

Question	Available Marks	Grade
1	12	
2	8	
3	10	
4	10	
5	10	
Total:	50	

Section A: Multiple Choice

Empl ID:

Question:	1	2	Total
Points:	12	8	20
Score:			

Question 1 [12 marks]

- a. If the Serial Peripheral Interface (SPI) protocol is configured to operate in SPI Mode 1, the logic level of the clock signal is: (2)
- ☐ low while idling
 - ☐ high while idling
 - ☐ always low
 - ☐ always alternating
- b. Loop Unrolling (LUR) is a common optimisation strategy that... (2)
- ☐ forces the processor to ignore loop condition checks to reduce overhead
 - ☐ parallelises iterations of a loop to improve runtime efficiency
 - ☐ removes unnecessary loop statements
 - ☐ reduces the number of iterations in a loop to reduce overhead
- c. Serial Peripheral Interface (SPI) and Inter-Integrated Circuit (I2C) each use a different means of selecting a slave device with which the master communicates. From the following options, choose the most accurate descriptions. (2)
- ☐ SPI and I2C both use a Slave Select line to initiate communication with a specific slave device.
 - ☐ With I2C, the master transmits a 7-bit address to indicate its slave device selection, whereas SPI uses a dedicated Slave Select line.
 - ☐ With SPI, the master transmits a 7-bit address to indicate its slave device selection, whereas I2C uses a dedicated Slave Select line.
 - ☐ SPI and I2C both use 7-bit addressing to indicate slave device selection.
- d. The STM32F051C6's non-volatile memory, also known as Flash memory, comprises the memory address space between... (2)
- ☐ 0x20000000 and 0x20001FFF
 - ☐ 0x08000000 and 0x08007FFF
 - ☐ 0x20000000 and 0x480017FF
 - ☐ 0x08000000 and 0x480017FF
- e. To facilitate RS-232-compliant communications with our STM board, the MAX232 chip converts our voltages from 0 V and 3.3 V, respectively, to... (2)
- ☐ +12 V and -12 V
 - ☐ -12 V and +12 V
 - ☐ 0 V and 12 V
 - ☐ -12 V and 0 V
- f. For which reason below is RISC being shown to be more popular than CISC for embedded systems development? (2)

- ☐ It does more processing in a clock cycle
- ☐ It is more cost-effective and more streamlined
- ☐ It gives better real-time performance
- ☐ It is easier to use and has more debugging support

Question 2 [8 marks]

Select either True or False for each of the following statements.

- a. The 7-bit data word 0b1110111 is to be transmitted with a parity bit appended; if using odd parity, the final 8-bit data word that would be transmitted is 0b11101111. (1)
☐ True ☐ False
- b. RS-422 allows for full-duplex communication between one transmitter and up to 32 receivers. (1)
☐ True ☐ False
- c. The FU (Function Unit) is considered to be a part of the ALU (Arithmetic Logic Unit). (1)
☐ True ☐ False
- d. An I2C slave sends a NACK bit when it is unable to receive/transmit because it is busy and thus not ready to start communication with the master. (1)
☐ True ☐ False
- e. A timer works by having the CNT register count up to the value in the PSC, starting from 0; then, when $CNT = PSC$, an overflow occurs and CNT is reset to 0. (1)
☐ True ☐ False
- f. The memory address 0x08000004 corresponds to the location of the Reset Vector. True or false? (1)
☐ True ☐ False
- g. In the general 3-stage CPU example discussed in class, the sequencing of the stages involved write-back as the first stage that gets executed in the sequence of actuating an instruction. (1)
☐ True ☐ False
- h. Thumb instructions can not only save space, but can also help increase overall program execution speeds compared to AArch32. (1)
☐ True ☐ False

Section B: Full QuestionsEmpl ID:

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Question:	3	4	5	Total
Points:	10	10	10	30
Score:				

Question 3 [10 marks]

- a. The ARM Cortex-M0 is said to have a three-stage pipeline. Draw a diagram to illustrate what this means, and briefly explain how it works.

(6)

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- b. Describe the terms "execution environment" and "runtime environment", and explain the relationship between them.

(4)

Question 4 [10 marks]

Imagine that you are developing firmware for an embedded system using a 32-bit microcontroller. The system utilises memory-mapped I/O to interface with an 8-bit ADC and a 32-bit timer.

The ADC provides the latest analog sample value at address 0xFFFF1000. The timer increments every 10 μs and is accessible at address 0xFFFF4000. Do not make any assumptions about the value at which the timer starts.

Write a C code snippet that determines the time it takes for the analog input signal to transition from a *trigger on* value – which for this program is set at a constant value of 200 – to a value below a *trigger off* value – in this program, use a constant value of 100.

Requirements:

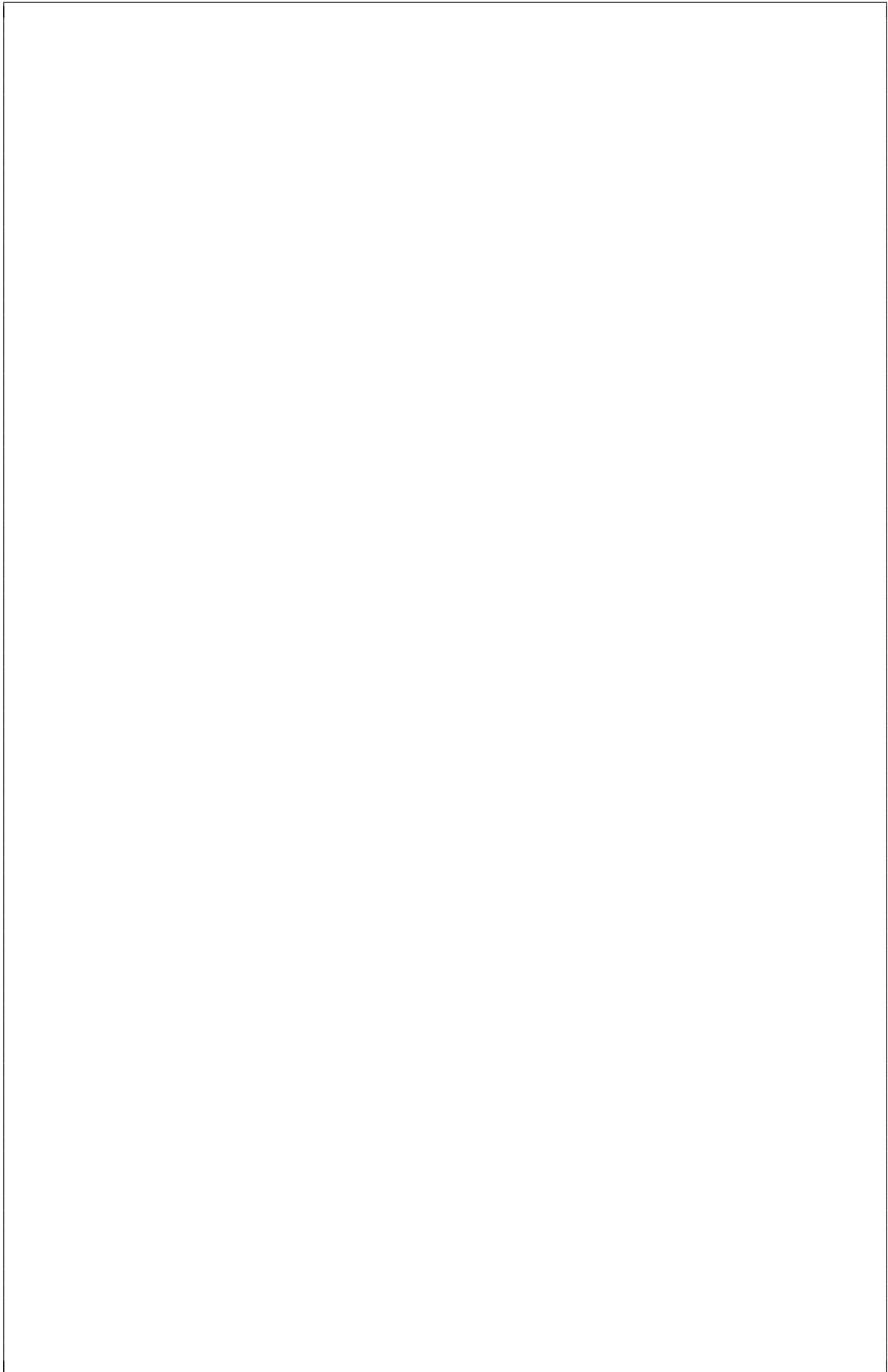
Use pointers to access the memory-mapped I/O addresses.

Calculate the time difference in microseconds.

Store the calculated time difference in a variable named *time_difference*.

Example run case:

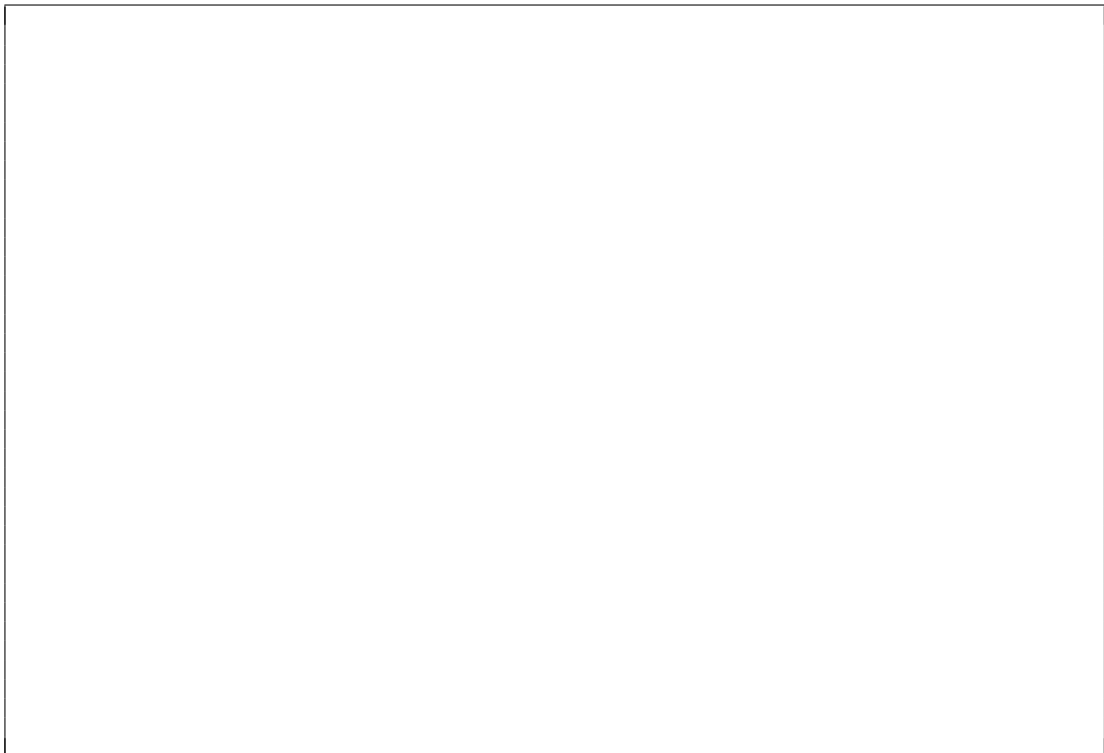
Here is an example case of running the program to help explain its operation. Consider that the analog input reaches 200 or above at time $t = 500 \mu\text{s}$, and then falls below 100 at $t = 5000 \mu\text{s}$; in this case, the *time_difference* should be 4500 μs .



**Question 5** [10 marks]

- a. Describe one benefit and one drawback to using RS-232 over I2C for embedded communications.

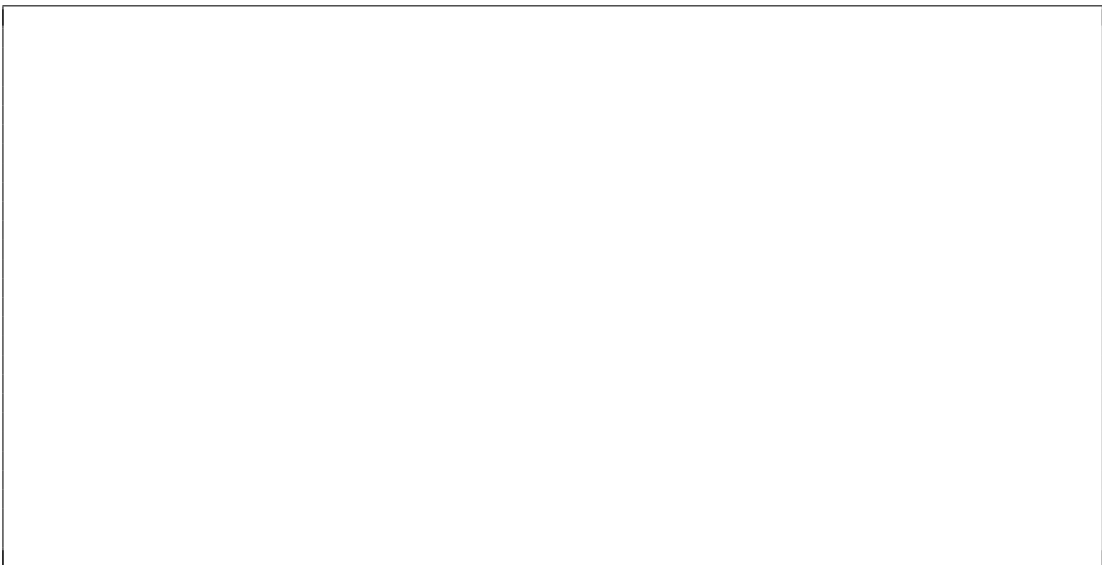
(2)

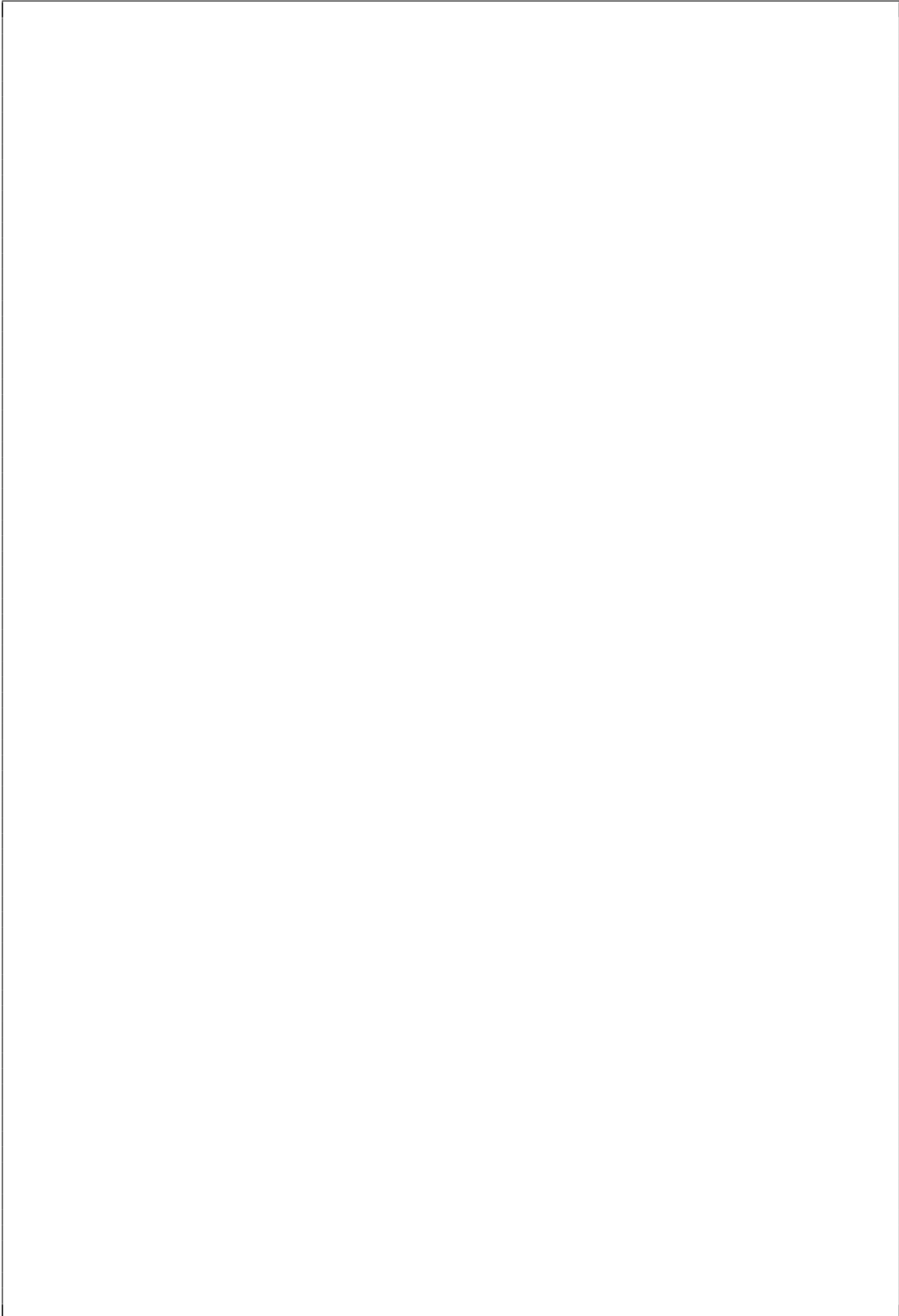


- b. When using SPI, is it necessary to have a common clock line between communicating devices? Why? And where would this signal be generated? (3)



- c. You have been asked to use RS-232 to transmit the data byte 0x4A to a receiving device. Assuming that Little Endian and even parity are being used, draw an appropriate timing diagram to illustrate this communication process. (5)





END OF TEST